Limitations and Uncertainty

> Part of the Functional Series. Every claim the platform makes has a limit. This document gathers them in one place, ranks where the uncertainty is largest, and maps which dashboard element carries which caveat. Read it before quoting any figure from the Findings section.

1. Four kinds of uncertainty

Kind	What it is	Example here
Measurement and data	The inputs are estimates or reconstructions	PRIMAP-hist before about 1970; land-use CO_2 (about ±2.6 $GtCO_2$ per year); the preliminary temperature dataset
Statistical	Noise in a fitted model	The 95% intervals on ETS forecasts and on the 0.486 slope
Structural	The model form may be wrong	A straight line through cumulative emissions; one fixed damping pattern per country
Scenario	The future depends on choices nobody has made	The 2% and 5% reductions; the held-flat land use; the rest-of-world share

The intervals the platform shows cover only the **statistical** kind. The other three do not appear as bands, which is why a narrow interval is not the same as a confident prediction.

2. Where the uncertainty is largest

Result	Main sources of uncertainty	Confidence
Historical emissions by country, since 1990	Inventory methods; the source choice (a few percent between sources)	High for trend and rank order
Global temperature anomaly	Preliminary dataset; baseline convention (a few hundredths of a degree)	High for the multi-decade trend; lower for the latest single year
Atmospheric CO_2 since 1959	Direct measurement	Very high
Emissions–temperature slope	Autocorrelation, non-CO_2 forcing, land-use estimates, non-constant relationship	Moderate: inside the IPCC range, but an analogue
ETS forecasts to 2043	Everything outside the past trend; five-year holdout is the only test	Low for levels beyond about a decade; better as a statement of "if trends continue"
Scenario pathways to 2040	The scenarios are assumptions, not forecasts	Comparative only
Implied temperature, 2040	All of the above, compounded, with only slope uncertainty shown	Comparative only

3. Limits specific to each stage

Data

- Territorial emissions only: emissions embodied in traded goods are not allocated to the consuming country.
- International aviation and shipping are included in the OWID World total but not in PRIMAP-hist or in country sums.
- The 40-country expanded set holds 92.2% of 2024 emissions; the other ~175 countries are in the rankings but not in the models.
- The latest year is the most likely to be revised.

Forecasting

- Univariate: each ETS forecast uses only that country's own CO_2 history. It does not know about policies, prices, energy transitions, wars or recessions.
- The test is five years long and contains the pandemic. Success there is not evidence of skill over twenty years.
- One model family (damped trend) is used for the long view; no model selection among others.
- Single-year data anomalies are flagged but not altered (Kuwait's 1991 oil fires distort its fit).

Relationship and scenarios

- The slope is estimated from a single observed history in which emissions, other gases and aerosols all rose together.
- Scenarios apply one percentage reduction to every country alike.
- Land use is held flat; rest-of-world follows the covered countries.
- Level figures in 2040 start from a five-year average anchor (1.32 °C), so they are only as good as that choice.

4. What the platform does to keep itself honest

- **Labels where the number is read.** Scenario results are marked "Illustrative · implied outcomes, not projections"; the relationship is marked "context, not proof"; the 1970-onward all-gas view is explicitly "not TCRE".
- **Figures come from the data, not from prose.** Numbers on the relationship and scenario pages are read from pipeline output, so a figure and its explanation cannot drift apart.
- **No attribution to countries.** Warming is never assigned to a country.
- **Failing safe.** If a source is stale or incomplete, the platform shows an explicit gap and a reason rather than a stale or partial number.
- **Disclosed, not reconciled.** Known differences between sources are stated and left visible.

5. Where the methodology appears in the dashboard

Where	What it carries
About page, "Methodology Summary"	A seven-row summary: dataset, countries, features, train/test split, models, forecasting, scenarios

Where	What it carries
Climate Correlation, "Methodology & sources"	Sources, coverage, vintage and licence for every dataset; the baseline used by each view; the temperature offset; why the emissions series differ; the two-totals note; the 1959 splice
Climate Correlation, "How this number was derived"	The seven-step derivation of the slope, including a "what this is not" step
Forecasts	The five-model comparison, the fitted ETS parameters, and the holdout overlay on each forecast
Scenario Comparison and **Overview**	The scenario definitions and the "illustrative" label on implied temperature

The About page is a short summary; the other pages hold the detail.

6. A checklist before quoting a number

1. Which **source** and which **scope** (gases, land use, international transport)?
2. Which **baseline year** or reference period?
3. Is it an **observation**, a **model fit**, a **forecast** or a **scenario**?
4. Does the uncertainty shown include **everything**, or only the statistical part?
5. Is it the **latest year** and so liable to revision?
6. Is it being used **comparatively** (fine) or as a **prediction** (usually not)?

See also

- Trend Analysis
- Forecasting
- The Emissions–Temperature Relationship
- Scenarios and Temperature Mapping
- Data Quality and Harmonization